

Randall Sim

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Education

Brown University

B.S. in Computer Science, GPA: 4.0

Providence, RI

Sep 2025 — May 2028

Stony Brook University

Computer Science, GPA: 4.0 (Transferred to Brown)

Stony Brook, NY

Aug 2024 — May 2025

Skills

Languages: Javascript/Typescript, Java, Python, Golang, Rust, C#

Frameworks: React, Next.js, Spring Boot, Express, Flask, .NET

Tools: AWS, Azure, Docker, Git, Linux, PostgreSQL, Redis, RabbitMQ

Work Experience

Microsoft

Redmond, WA

Software Engineer Intern

May 2026 — Aug 2026

- Shipped AI feature for Intune Copilot that converts natural-language device queries into interactive data visualizations; deployed to selfhost and released to Intune's **private preview** enterprise customers.
- Removed a **50K-device rendering limit** by rewriting KQL queries to aggregate server-side, validated against Microsoft IT's 2M+ device fleet.
- Developed the full stack: a React graph-editing interface and a C# GPT skill on Microsoft's Security Copilot platform, integrated into the existing Intune portal.
- Added click telemetry to the AI skill popup and graph editor to measure adoption and tune graph generation defaults.

Stony Brook Campus Residences

Stony Brook, NY

Software Engineer Intern

Sep 2024 — May 2025

- **Rebuilt PrintStats**, Stony Brook's official printer dashboard. Migrated the entire C# .NET backend to Express.js, utilizing asynchronous functions to achieve a **12x improvement (6 min → 30 sec)** in loading printer metadata via SNMP.
- Built and maintained dashboards (C# .NET, Express, React, SQL) used by 100+ professional staff to monitor campus Scalas (TVs), computer closets, and KACE-managed endpoints.
- Modernized CI/CD by replacing manual Remote Desktop + IIS setup with GitHub Actions and Windows dev containers.

Personal Projects

epsilon.stuysu.org

- Founded Epsilon, an all-in-one platform serving **3,300+ daily users** and **400+ student organizations** at Stuyvesant HS—sole engineer for first year, now maintained by the school.
- Designed a PostgreSQL database with role-based permissions across **100+ security policies** and **30+ schema migrations**; integrated additional tools (calendar, emails, room reservations) via 20+ Deno edge functions.
- Deployed a React frontend, with extensive modules (club catalog, attendance, valentines) and dashboards (admin, organization, user).

internships.dev

- Co-founded internships.dev, a high-speed job alerts platform serving **100+ users**, tracking **2000+ tech internships** across **200+ job boards** with sub-5-minute alert delivery.
- Engineered a fault-tolerant distributed web scraping system (Python, Redis), scraping **60,000+ pages/day across 5 worker nodes**.
- Designed an abstract scraper interface and built an AI pipeline (Claude + Playwright MCP) to automate site-specific implementations.
- Created an alerts microservice with Node.js and RabbitMQ to deliver notifications to Discord, GitHub, and email.
- Developed backend REST API with **Spring Boot**, including auth services, web scraper pipelines, and a Stripe subscription service.

[Semantic Fetch](#)

- Built a **Rust** AI agent (rig-rs) and **TypeScript SDK** that routes natural-language queries to REST endpoints, with a client-side cache reducing repeated-query latency to match direct API calls.
- Demonstrated the SDK in a full-stack reference app (**Go** REST backend, **Next.js** frontend) showing natural-language routing integrated into a real CRUD application.

Relevant Coursework

Data Structures & Algorithms, Programming Abstractions, Systems Programming, Analysis of Algorithms, Fundamentals of AI, Software Engineering, Computer Networking, Applied Cryptography, Discrete Math, Calculus I, II, and III.